Solution Design Document

Platform LMS RENJANA - Fase 2

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1. Arsitektur Sistem

1.1 High-Level Architecture

```
┌─────────────────────────────────────────────────────────────────┐
│                          CLIENT LAYER                           │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐           │
│  │   Desktop    │  │   Mobile     │  │   Tablet     │           │
│  │   Browser    │  │   Browser    │  │   Browser    │           │
│  └──────┬───────┘  └──────┬───────┘  └──────┬───────┘           │
└─────────┼─────────────────┼─────────────────┼───────────────────┘
          │                 │                 │
          └─────────────────┴─────────────────┘
                            │
                            │ HTTPS
                            ▼
┌─────────────────────────────────────────────────────────────────┐
│                       APPLICATION LAYER                         │
│  ┌──────────────────────────────────────────────────────────┐   │
│  │                     Next.js 16 (Full-stack)              │   │
│  │  ┌──────────────┐  ┌──────────────┐  ┌──────────────────┐  │   │
│  │  │   Pages/     │  │    API       │  │   Server Actions  │  │   │
│  │  │   Routes     │  │   Routes     │  │   (Server-side)   │  │   │
│  │  └──────────────┘  └──────────────┘  └──────────────────┘  │   │
│  │                                                            │   │
│  │  ┌──────────────┐  ┌──────────────┐  ┌──────────────────┐  │   │
│  │  │  NextAuth    │  │   Prisma     │  │   File Upload     │  │   │
│  │  │   (Auth)     │  │   (ORM)      │  │   Service         │  │   │
│  │  └──────────────┘  └──────────────┘  └──────────────────┘  │   │
│  └──────────────────────────────────────────────────────────┘   │
└─────────────────────────────────────────────────────────────────┘
                            │
```

```
                        ▼
┌──────────────────────────────────────────────────────────────┐
│                        DATA LAYER                            │
│  ┌────────────────┐  ┌────────────────┐  ┌──────────────────┐  │
│  │   PostgreSQL   │  │   Cloudinary   │  │   Email Service  │  │
│  │   (Database)   │  │   (File Storage)│  │   (Resend)       │  │
│  └────────────────┘  └────────────────┘  └──────────────────┘  │
└──────────────────────────────────────────────────────────────┘
```

1.2 Component Architecture

```
┌──────────────────────────────────────────────────────────────┐
│                      FRONTEND COMPONENTS                     │
├──────────────────────────────────────────────────────────────┤
│                                                              │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐        │
│  │   Landing    │  │   Events     │  │  Dashboard   │        │
│  │    Page      │  │   Module     │  │   Module     │        │
│  └──────────────┘  └──────────────┘  └──────────────┘        │
│                                                              │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐        │
│  │    Admin     │  │ Instructor   │  │   Finance    │        │
│  │    Panel     │  │   Panel      │  │   Panel      │        │
│  └──────────────┘  └──────────────┘  └──────────────┘        │
│                                                              │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐        │
│  │ Attendance   │  │ Certificate  │  │   Shared     │        │
│  │   Module     │  │   Module     │  │ Components   │        │
│  └──────────────┘  └──────────────┘  └──────────────┘        │
│                                                              │
└──────────────────────────────────────────────────────────────┘
```

---

2. Tech Stack

2.1 Frontend

Technology	Version	Purpose
Next.js	16.x	React Framework (Full-stack)
React	19.x	UI Library
Tailwind CSS	4.x	Styling
Framer Motion	12.x	Animations
Radix UI	Latest	Accessible Components
Lucide Icons	Latest	Icon Library

2.2 Backend

Technology	Version	Purpose
Next.js API Routes	16.x	REST API endpoints
Server Actions	16.x	Server-side mutations

Prisma	5.x	ORM & Database toolkit
NextAuth.js	5.x	Authentication

2.3 Database & Services

Technology	Purpose
PostgreSQL 15	Primary Database
Cloudinary	Image & File Storage
Resend	Email Service

2.4 DevOps

Technology	Purpose
Docker	Containerization
Docker Compose	Local Development
GitHub Actions	CI/CD
Vercel / VPS	Hosting

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3. Database Design

3.1 Entity Relationship Diagram

```
┌──────────────────┐        ┌──────────────────┐        ┌──────────────────┐
│       User       │        │      Course      │        │       Event      │
├──────────────────┤        ├──────────────────┤        ├──────────────────┤
│ id (PK)          │        │ id (PK)          │        │ id (PK)          │
│ email            │        │ title            │        │ title            │
│ password         │        │ description      │        │ description      │
│ name             │        │ thumbnail        │        │ date             │
│ role             │        │ instructorId(FK) │        │ location         │
│ avatar           │        │ status           │        │ quota            │
│ createdAt        │        │ createdAt        │        │ flyerUrl         │
│ updatedAt        │        │ updatedAt        │        │ status           │
└──────────┬───────┘        └──────────┬───────┘        └──────────┬───────┘
           │                           │                           │
           │    ┌──────────────────────┼───────────────────────────┤
           │    │                      │                           │
           ▼    ▼                      ▼                           ▼

┌──────────────────┐        ┌──────────────────┐        ┌──────────────────┐
│   Enrollment     │        │   Attendance     │        │  Registration    │
├──────────────────┤        ├──────────────────┤        ├──────────────────┤
│ id (PK)          │        │ id (PK)          │        │ id (PK)          │
│ userId (FK)      │        │ userId (FK)      │        │ userId (FK)      │
│ courseId (FK)    │        │ eventId (FK)     │        │ eventId (FK)     │
│ progress         │        │ status           │        │ status           │
│ status           │        │ checkedInAt      │        │ registeredAt     │
│ enrolledAt       │        │ createdAt        │        │ createdAt        │
└──────────────────┘        └──────────────────┘        └──────────────────┘
```

```
        │                       │
        │                       │
        ▼                       ▼
┌──────────────────────────────────────────┐
│                Certificate                 │
├──────────────────────────────────────────┤
│ id (PK)                                   │
│ certificateNumber (Unique)                │
│ userId (FK)                               │
│ eventId (FK) / courseId (FK)              │
│ type (COURSE / EVENT)                     │
│ pdfUrl                                    │
│ issuedAt                                  │
│ verificationCode                          │
└──────────────────────────────────────────┘
```

3.2 Prisma Schema

```
// prisma/schema.prisma

generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

enum UserRole {
  LEARNER
  INSTRUCTOR
  ADMIN
  MANAGER
  FINANCE
}

enum EventStatus {
  DRAFT
  PUBLISHED
  COMPLETED
  CANCELLED
}

enum AttendanceStatus {
  REGISTERED
  PRESENT
  ABSENT
  LATE
}

model User {
  id            String         @id @default(cuid())
  email         String         @unique
  password      String
```

```
  name          String
  role          UserRole       @default(LEARNER)
  avatar        String?
  createdAt     DateTime       @default(now())
  updatedAt     DateTime       @updatedAt

  enrollments   Enrollment[]
  registrations Registration[]
  attendances   Attendance[]
  certificates  Certificate[]
  instructorOf  Course[]       @relation("Instructor")
}

model Course {
  id           String       @id @default(cuid())
  title        String
  description  String
  thumbnail    String?
  instructorId String
  instructor   User         @relation("Instructor", fields: [instructorId], references: [id])
  status       String       @default("draft")
  createdAt    DateTime     @default(now())
  updatedAt    DateTime     @updatedAt

  enrollments  Enrollment[]
  certificates Certificate[]
}

model Event {
  id                   String        @id @default(cuid())
  title                String
  description          String
  date                 DateTime
  endDate              DateTime?
  location             String
  isOnline             Boolean       @default(false)
  meetingUrl           String?
  quota                Int
  flyerUrl             String?
  materials            String[]
  status               EventStatus   @default(DRAFT)
  registrationDeadline DateTime?
  createdAt            DateTime      @default(now())
  updatedAt            DateTime      @updatedAt

  registrations Registration[]
  attendances   Attendance[]
  certificates  Certificate[]
}

model Enrollment {
  id         String   @id @default(cuid())
  userId     String
  user       User     @relation(fields: [userId], references: [id])
  courseId   String
  course     Course   @relation(fields: [courseId], references: [id])
```

```
  progress   Int      @default(0)
  status     String   @default("active")
  enrolledAt DateTime @default(now())

  @@unique([userId, courseId])
}

model Registration {
  id           String   @id @default(cuid())
  userId       String
  user         User     @relation(fields: [userId], references: [id])
  eventId      String
  event        Event    @relation(fields: [eventId], references: [id])
  status       String   @default("pending")
  registeredAt DateTime @default(now())

  @@unique([userId, eventId])
}

model Attendance {
  id          String           @id @default(cuid())
  userId      String
  user        User             @relation(fields: [userId], references: [id])
  eventId     String
  event       Event            @relation(fields: [eventId], references: [id])
  status      AttendanceStatus @default(REGISTERED)
  checkedInAt DateTime?
  createdAt   DateTime         @default(now())

  @@unique([userId, eventId])
}

model Certificate {
  id                String   @id @default(cuid())
  certificateNumber String   @unique
  userId            String
  user              User     @relation(fields: [userId], references: [id])
  eventId           String?
  event             Event?   @relation(fields: [eventId], references: [id])
  courseId          String?
  course            Course?  @relation(fields: [courseId], references: [id])
  type              String   // "COURSE" or "EVENT"
  pdfUrl            String?
  verificationCode  String   @unique
  issuedAt          DateTime @default(now())
}
```

---

4. API Design

4.1 API Endpoints

Authentication

Method	Endpoint	Description
POST	`/api/auth/register`	User registration

POST	`/api/auth/login`	User login
POST	`/api/auth/logout`	User logout
GET	`/api/auth/me`	Get current user

Events

Method	Endpoint	Description
GET	`/api/events`	List all published events
GET	`/api/events/[id]`	Get event detail
POST	`/api/events`	Create event (Admin)
PUT	`/api/events/[id]`	Update event (Admin)
DELETE	`/api/events/[id]`	Delete event (Admin)

Registrations

Method	Endpoint	Description
POST	`/api/events/[id]/register`	Register to event
DELETE	`/api/events/[id]/register`	Cancel registration
GET	`/api/events/[id]/registrations`	List registrations (Admin)
GET	`/api/events/[id]/registrations/export`	Export registrations to Excel (Admin)

Attendance

Method	Endpoint	Description
POST	`/api/events/[id]/checkin`	Check-in via QR
POST	`/api/events/[id]/checkin/[userId]`	Manual check-in (Admin)
GET	`/api/events/[id]/attendance`	Get attendance list
GET	`/api/events/[id]/attendance/export`	Export attendance to Excel (Admin)

Certificates

Method	Endpoint	Description
GET	`/api/certificates`	User's certificates
GET	`/api/certificates/[id]/download`	Download PDF
GET	`/api/certificates/verify/[code]`	Verify certificate

4.2 API Response Format

```
// Success Response
{
  "success": true,
  "data": { ... },
```

```
  "message": "Operation successful"
}

// Error Response
{
  "success": false,
  "error": {
    "code": "ERROR_CODE",
    "message": "Human readable message"
  }
}

// Paginated Response
{
  "success": true,
  "data": [ ... ],
  "pagination": {
    "page": 1,
    "limit": 10,
    "total": 100,
    "totalPages": 10
  }
}
```

---

5. UI/UX Design

5.1 Design System

Color Palette

Token	Light Mode	Dark Mode	Usage
Primary	`#2563eb`	`#3b82f6`	Buttons, Links, Accents
Background	`#ffffff`	`#0f172a`	Page background
Surface	`#f8fafc`	`#1e293b`	Cards, Modals
Text	`#1e293b`	`#f1f5f9`	Body text
Success	`#22c55e`	`#22c55e`	Success states
Error	`#ef4444`	`#ef4444`	Error states

Typography

Element	Font	Size	Weight
H1	Inter	48px	800
H2	Inter	36px	700
H3	Inter	24px	600
Body	Inter	16px	400
Small	Inter	14px	400

5.2 Page Wireframes

Events List Page

```
┌──────────────────────────────────────────────────────────────────────┐
│  🏛️ RENJANA                    [Events] [Courses] [Login] [🌙]      │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  Upcoming Events                                                     │
│  ═══════════════                                                     │
│                                                                      │
│  [🔍 Search events...]                    [Filter: All ▼]            │
│                                                                      │
│  ┌──────────────────┐  ┌──────────────────┐  ┌──────────────────┐   │
│  │ ┌──────────────┐ │  │ ┌──────────────┐ │  │ ┌──────────────┐ │   │
│  │ │    FLYER     │ │  │ │    FLYER     │ │  │ │    FLYER     │ │   │
│  │ │    IMAGE     │ │  │ │    IMAGE     │ │  │ │    IMAGE     │ │   │
│  │ └──────────────┘ │  │ └──────────────┘ │  │ └──────────────┘ │   │
│  │                  │  │                  │  │                  │   │
│  │ Webinar Mediasi  │  │ Workshop Hukum   │  │ Pelatihan ADR    │   │
│  │                  │  │                  │  │                  │   │
│  │ 📅 30 Jan 2026   │  │ 📅 15 Feb 2026   │  │ 📅 28 Feb 2026   │   │
│  │ 📍 Jakarta       │  │ 📍 Online        │  │ 📍 Surabaya      │   │
│  │ 👥 50/100        │  │ 👥 80/100        │  │ 👥 20/50         │   │
│  │                  │  │                  │  │                  │   │
│  │ [  View Detail  ]│  │ [  View Detail  ]│  │ [  View Detail  ]│   │
│  └──────────────────┘  └──────────────────┘  └──────────────────┘   │
│                                                                      │
│  ────────────────────────────────────────────────────────────────    │
│  [  1  ] [ 2 ] [ 3 ] [ > ]                                           │
│                                                                      │
└──────────────────────────────────────────────────────────────────────┘
```

Event Detail Page

```
┌──────────────────────────────────────────────────────────────────────┐
│  ← Kembali ke Events                                                 │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  ┌────────────────────────┐  ┌──────────────────────────────────┐   │
│  │                        │  │                                  │   │
│  │                        │  │  Webinar: Mediasi Hukum Bisnis   │   │
│  │                        │  │  ═══════════════════════════     │   │
│  │     EVENT FLYER        │  │                                  │   │
│  │       IMAGE            │  │  📅 Sabtu, 30 Januari 2026       │   │
│  │                        │  │  🕐 09:00 - 12:00 WIB            │   │
│  │                        │  │  📍 Hotel Grand Hyatt, Jakarta   │   │
│  │                        │  │  👥 Kuota: 50/100 peserta        │   │
│  │                        │  │                                  │   │
│  └────────────────────────┘  │  ┌────────────────────────┐      │   │
│                              │  │                        │      │   │
│                              │  │  [  🎫 DAFTAR SEKARANG  ]      │   │
│                              │  │                        │      │   │
│                              │  └────────────────────────┘      │   │
│                              └──────────────────────────────────┘   │
│                                                                      │
```

```
│  ┌──────────────────────────────────────────────────────────┐  │
│  │  Deskripsi                                                │  │
│  │  ══════════                                               │  │
│  │                                                           │  │
│  │  Webinar ini akan membahas teknik-teknik mediasi dalam    │  │
│  │  penyelesaian sengketa bisnis. Peserta akan mendapatkan...  │  │
│  └──────────────────────────────────────────────────────────┘  │
│                                                                  │
│  ┌──────────────────────────────────────────────────────────┐  │
│  │  Materi                                                   │  │
│  │  ══════                                                   │  │
│  │                                                           │  │
│  │  📄 Panduan Mediasi.pdf                  [ Download ]      │  │
│  │  📄 Slide Presentasi.pptx                [ Download ]      │  │
│  └──────────────────────────────────────────────────────────┘  │
│                                                                  │
└──────────────────────────────────────────────────────────────────┘
```

Admin Check-in Dashboard

```
┌──────────────────────────────────────────────────────────────────┐
│  🏛 RENJANA ADMIN          [Dashboard] [Events] [Users] [Logout] │
├──────────────────────────────────────────────────────────────────┤
│                                                                  │
│  Event: Webinar Mediasi Hukum Bisnis                             │
│  ═══════════════════════════════                                 │
│                                                                  │
│  ┌──────────────┐  ┌──────────────────────────────────┐         │
│  │              │  │        STATISTIK KEHADIRAN        │         │
│  │  ▀▀▀▀▀▀▀▀▀   │  │  ────────────────────────────────  │         │
│  │  ▀▀ QR   ▀▀  │  │                                   │         │
│  │  ▀▀ CODE ▀▀  │  │  ✅ Hadir      : 35 peserta       │         │
│  │  ▀▀▀▀▀▀▀▀▀   │  │  ⏳ Belum      : 15 peserta       │         │
│  │              │  │  ❌ Tidak Hadir: 0 peserta        │         │
│  │  [ Print QR ] │  │                                   │         │
│  │              │  │  Progress: ▓▓▓
```

Certificate Template

```
┌──────────────────────────────────────────────────────────────────┐
│                                                                  │
│                          ═══════════                             │
│                            SERTIFIKAT                            │
│                          ═══════════                             │
│                                                                  │
│                      🏛️ RENJANA Learning                         │
│                                                                  │
│                        dengan ini menyatakan                     │
│                                                                  │
│                    ═══════════════════                           │
│                        AHMAD SUBARJO                             │
│                    ═══════════════════                           │
│                                                                  │
│                  telah berhasil mengikuti kegiatan               │
│                                                                  │
│                    "Webinar Mediasi Hukum Bisnis"                │
│                                                                  │
│                  yang diselenggarakan pada tanggal               │
│                        30 Januari 2026                           │
│                                                                  │
│                                                                  │
│   Jakarta, 30 Januari 2026                                       │
│                                                                  │
│                                                                  │
│   ________________                      ________________         │
│   Direktur Renjana                      Ketua Panitia            │
│                                                                  │
│                                                                  │
│   ┌─────────┐                                                    │
│   │ QR CODE │   No. Sertifikat: RJN-2026-00001                   │
│   └─────────┘   Verifikasi: renjana.com/verify/ABC123            │
│                                                                  │
└──────────────────────────────────────────────────────────────────┘
```

---

6. Security Design

6.1 Authentication Flow

```
┌──────────┐                    ┌──────────┐                ┌──────────┐
│  Client  │                    │  Server  │                │ Database │
└────┬─────┘                    └────┬─────┘                └────┬─────┘
     │                               │                           │
     │  1. POST /login               │                           │
     │  {email, password}            │                           │
     │──────────────────────────────>│                           │
     │                               │  2. Find user by email    │
     │                               │──────────────────────────>│
     │                               │                           │
     │                               │  3. Return user           │
     │                               │<──────────────────────────│
     │                               │                           │
     │                               │  4. Verify password       │
```

```
    |                              |      (bcrypt.compare)     |
    |                              |                           |
    |  5. Set session cookie       |                           |
    |  (HTTP-only, Secure)         |                           |
    |<-----------------------------|                           |
    |                              |                           |
```

6.2 Security Measures

Area	Measure
Password	bcrypt hash (12 rounds)
Session	HTTP-only, Secure, SameSite cookies
API	Rate limiting (100 req/min)
File Upload	Type validation, max size 10MB
Input	Server-side validation (Zod)
CSRF	Built-in NextAuth protection

---

7. Integration Design

7.1 File Upload (Cloudinary)

```
// lib/cloudinary.ts
import { v2 as cloudinary } from 'cloudinary';

cloudinary.config({
  cloud_name: process.env.CLOUDINARY_CLOUD_NAME,
  api_key: process.env.CLOUDINARY_API_KEY,
  api_secret: process.env.CLOUDINARY_API_SECRET,
});

export async function uploadFile(file: File, folder: string) {
  const bytes = await file.arrayBuffer();
  const buffer = Buffer.from(bytes);

  return new Promise((resolve, reject) => {
    cloudinary.uploader.upload_stream(
      { folder: `renjana/${folder}` },
      (error, result) => {
        if (error) reject(error);
        else resolve(result);
      }
    ).end(buffer);
  });
}
```

7.2 Email (Resend)

```
// lib/email.ts
import { Resend } from 'resend';
```

```
const resend = new Resend(process.env.RESEND_API_KEY);

export async function sendRegistrationConfirmation(to: string, event: Event) {
  await resend.emails.send({
    from: 'RENJANA <noreply@renjana.com>',
    to,
    subject: `Konfirmasi Pendaftaran: ${event.title}`,
    html: `<h1>Pendaftaran Berhasil!</h1>...`
  });
}
```

7.3 PDF Generation (Certificate)

```
// lib/certificate.ts
import PDFDocument from 'pdfkit';

export async function generateCertificate(data: CertificateData) {
  const doc = new PDFDocument({ size: 'A4', layout: 'landscape' });

  // Add certificate content
  doc.fontSize(36).text('SERTIFIKAT', { align: 'center' });
  doc.fontSize(24).text(data.userName, { align: 'center' });
  doc.fontSize(16).text(`"${data.eventTitle}"`, { align: 'center' });

  // Add QR code
  const qrCode = await generateQRCode(data.verificationUrl);
  doc.image(qrCode, 50, 350, { width: 80 });

  return doc;
}
```

---

8. Deployment Architecture

8.1 Production Environment

```
┌──────────────────────────────────────────────────────────────────┐
│                        PRODUCTION ENVIRONMENT                    │
├──────────────────────────────────────────────────────────────────┤
│                                                                  │
│  ┌──────────────────┐                                            │
│  │   Cloudflare     │  CDN + DDoS Protection                     │
│  │   (DNS + CDN)    │                                            │
│  └──────────┬───────┘                                            │
│             │                                                    │
│             ▼                                                    │
│  ┌──────────────────┐                                            │
│  │   Vercel /       │  Next.js Hosting                           │
│  │   VPS (Docker)   │  Auto-scaling                              │
│  └──────────┬───────┘                                            │
│             │                                                    │
│             ├──────────────────┬──────────────────┐              │
│             │                  │                  │              │
│             ▼                  ▼                  ▼              │
```

```
│  ┌──────────────────┐ ┌──────────────────┐ ┌──────────────────┐    │
│  │  PostgreSQL      │ │   Cloudinary     │ │     Resend       │    │
│  │  (Supabase/      │ │   (Files)        │ │     (Email)      │    │
│  │   Neon/Railway)  │ │                  │ │                  │    │
│  └──────────────────┘ └──────────────────┘ └──────────────────┘    │
│                                                                     │
└─────────────────────────────────────────────────────────────────────┘
```

8.2 Docker Configuration

```
# Dockerfile
FROM node:20-alpine AS base

FROM base AS deps
WORKDIR /app
COPY package*.json ./
RUN npm ci

FROM base AS builder
WORKDIR /app
COPY --from=deps /app/node_modules ./node_modules
COPY . .
RUN npx prisma generate
RUN npm run build

FROM base AS runner
WORKDIR /app
ENV NODE_ENV production
COPY --from=builder /app/public ./public
COPY --from=builder /app/.next/standalone ./
COPY --from=builder /app/.next/static ./.next/static

EXPOSE 3000
CMD ["node", "server.js"]
```

---

9. Estimasi Effort

9.1 Sprint Breakdown

Sprint	Durasi	Deliverable	Effort
Sprint 1	2 minggu	Database + Auth + API Foundation	80 jam
Sprint 2	2 minggu	Event Module (CRUD, Registration)	80 jam
Sprint 3	2 minggu	Attendance + Certificate	80 jam
UAT	2 minggu	Testing + Bug Fixing	60 jam
Go-Live	1 minggu	Deployment + Training	40 jam

9.2 Resource Allocation

Role	Allocation	Responsibility

Full-stack Developer	100%	Development, API, Database
UI/UX Designer	25%	Design review, Asset creation
QA Engineer	50% (Sprint UAT)	Testing
Project Manager	20%	Coordination

9.3 Dependencies

Item	Status	Owner	Deadline
PostgreSQL Server	Pending	IT Ops	Sprint 1
Cloudinary Account	Pending	IT	Sprint 1
Resend Account	Pending	IT	Sprint 2
Certificate Template Design	Pending	Design	Sprint 3
Domain SSL	Done	IT Ops	-

---

10. Sign-Off

Role	Nama	Tanda Tangan	Tanggal
IT Lead	_____	_____	_____
Business Lead	_____	_____	_____
Project Manager	_____	_____	_____

Dokumen ini merupakan deliverable Fase 3: Solution Design

Versi: 1.0 | Status: Draft for Review